Introduction to Probability

Why Probability?

In our day to day life, a lot of random & uncertain events happens & to predict that we use probability.

Ex- Launching a business, weather forecast, etc.

What is Probability?

It is a way to measure how likely an event is going to occur.

Formula → Let A be an event, then

$$P(A) = \frac{\text{No. of favourable outcomes}}{\text{Total No. of outcomes}}$$

Ex-

(1) Tossing a coin → 2 faces → H, T

Here, event → Getting a head

So,

$$P(H) = \frac{1}{2}$$

(2) Rolling a Die → 6 faces → 1, 2, 3, 4, 5, 6

Here, event → Getting a 2

So,

$$P(2) = \frac{1}{6}$$

(3) Drawing a card → 52 cards

Drawing a card : 52 cards

Here, event → drawing a black king

52
↓
13 diamonds ●
13 Spade ●
13 Clubs ●
13 Heart ●

So,

$$P(\text{Black King}) = \frac{2}{52}$$

or

$$\boxed{P(\text{Black King}) = \frac{1}{26}}$$

Questions :-

(1) Prob. of drawing an Ace → $\frac{\cancel{4}}{\cancel{52}} = \frac{1}{13}$

(2) Prob. of getting a tail → $\frac{1}{2}$

(3) Getting an even no. on rolling a die → $\frac{\cancel{3}}{\cancel{6}} = \frac{1}{2}$

(4) Getting a no. greater than 2 on rolling a die → $\frac{\cancel{4}}{\cancel{6}} = \frac{2}{3}$

* Random Experiment :-

It is an event that occurs where the exact outcome cannot be predicted.

Ex- Flipping a coin, Rolling a die, etc.

Trial :- Repetition of an experiment along with recording outcomes

Outcome :- Result of a single trial.

Sample Space :- Set of all possible outcomes

Ex - Toss a coin → {H, T}

Sample space :- Set of all possible outcomes

Ex → Tossing a coin → 2 → {H, T}

Rolling a die → 6 → {1, 2, 3, 4, 5, 6}

Tossing 2 coins → 4 → {HH, TT, HT, TH}

Event :- It is a subset of sample space in a random experiment.

Ex → Tossing a coin → Possible Outcomes → 2

(Tossing a coin = experiment)

Sample space → {H, T}

Event → {H} (subset of sample space)

Compound Event :- Collection of simple events grouped together on the basis of a specific criteria

Ex- Getting an odd no. > 2 on rolling a die

Complement of an event :-

Consists of all outcomes that are not present in the subset of sample space. It is denoted by A^c

Ex- Let event $A = \{2, 4, 6\}$

A^c = not A = $\{1, 3, 5\}$ } Die example

Union of Events :-

It includes all the possible outcomes that are present at least once in an event. It is denoted by $\cup$

Ex- Let A & B are two events

Union → $A \cup B$

$A = \{1, 2, 3, 5\}$; $B = \{2, 4, 5, 7, 8\}$

$$A \cup B = \{1, 2, 3, 4, 5, 7, 8\}$$

Intersection of Events :-

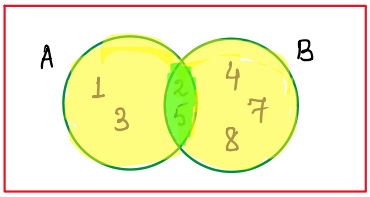
It include all possible outcomes that are common in between the events. It is denoted by $\cap$.

Ex- Let A & B are two events

Intersection $\rightarrow$ $A \cap B$

$A = \{1, 2, 3, 5\}$; $B = \{2, 4, 5, 7, 8\}$

$A \cap B = \{2, 5\}$



● → union
● → intersection

Ex- On rolling a die → Let TWO events be-

Event A = Getting an odd no.

Event B = Getting a no. greater than 2

Subset for A → $\{1, 3, 5\}$ ⇝ 3/6

" " B → $\{3, 4, 5, 6\}$ ⇝ 4/6

① $A \cup B = \{1, 3, 4, 5, 6\}$ ⇝ 5/6

② $A \cap B = \{3, 5\}$ ⇝ 2/6

★ Mutually Exclusive Events :-

- These events cannot occur simultaneously
- If one event happens, the other event becomes impossible.

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Ex- Getting a 2 or 5 at the same time on rolling a die

Getting H or T together at the same toss

∴ Let A & B are 2 events, then

$$P(A \cap B) = 0$$

* Dependent Events :-

When the occurrence of 1 event affects the occurrence of 2nd event, then these are known as dependent events.

Ex- 1st chance → Drawing an Ace from a pack of cards

$$P(A) = \frac{4}{52}$$ { card is kept aside }

2nd chance → $P(A) = \frac{3}{51}$

* Independent Events :-

One scenario does not impact the other scenario.

Ex- Tossing 1 coin two times

1st toss → Tossing a Head → 1/2

2nd toss → Tossing a Head → 1/2

Difference b/w Independent & Mutually Exclusive Events :-

Independent

One event does not affect the other

Mutually Exclusive

Cannot occur together